Stark resonances under exponential-cosine screening: through-threshold maps for the quantum-plasma hydrogen atom

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The exponential-cosine screened Coulomb (ECSC) potential of dense quantum plasmas differs from Debye screening in one structural respect: its tail oscillates. We compute the resonance spectrum of the hydrogen atom under ECSC screening combined with a static electric field with a coupled-channel complex-scaling method, exploiting an exact identity — the ECSC potential is the half-sum of two Yukawa potentials at complex screening parameters — which also yields the analytic rotation bound $\theta_c < \pi/4$. Three results organize the physics. At zero field, parametric continuation addresses the intermediate screening window left open in the literature: within the sector that uniform complex scaling can expose, no isolated ground pole survives beyond the critical screening $\lambda_c = 0.7205$ — the pole merges into the rotated continuum, consistent with the conjectured virtual-state scenario — while the $2p$ pole detaches at $\lambda_c(2p) = 0.148205$ (all digits confirmed by a recent 60-digit benchmark) as a genuine resonance, rotation-stable to 10^{-13}. With the field on, we map the ground resonance on a 5×19 grid of the (F, λ) plane: at every field strength the resonance crosses $E_r = 0$ near $\lambda^* \approx 0.68$ — nearly field-independent, a corollary of the vanishing field displacement at threshold — and continues smoothly as a positive-energy resonance, a regime the companion Debye map only approaches near its much larger critical screening. Finally, comparing ECSC and Debye screening at matched binding energy isolates the tail effect on ionization: the cosine factor extinguishes the assisting Coulombic tail well before its first zero, making the Stark barrier higher and wider, and suppresses the field-ionization width by roughly a factor of two at weak field — a suppression that fades as the field strengthens and the barrier peak moves inward to where the two potentials coincide.

I. INTRODUCTION

Screening in a dense, degenerate electron gas is not exponential. Quantum hydrodynamic treatments of test-charge screening in quantum plasmas [1] lead to the exponential-cosine screened Coulomb (ECSC) potential,

$$V(r) = -\frac{Z}{r}\, e^{-\lambda r} \cos(\lambda r), \tag{1}$$

whose study as a model interaction long predates that derivation [2]. The oscillating factor is not a cosmetic difference from the Debye–Hückel form: the potential changes sign, its tail alternates between attraction and repulsion, and this modifies both the bound spectrum — critical screening parameters differ substantially [3] — and the resonance structure. Field-free ECSC resonances have recently been mapped with high precision by the complex-scaling pseudospectral work of Jiao and collaborators [4], who located the loss of the ground state at $\lambda_c = 0.720524$ and the emergence of well-defined s-wave shape resonances at $\lambda_{r+} = 0.90035$, and who explicitly note that the behavior in the intermediate window is unexplained, conjecturing conversion into virtual states. Perturbative field responses (polarizabilities and hyperpolarizabilities) of ECSC-screened atoms are likewise known [5], and the one nonperturbative Stark calculation under ECSC screening — complex-scaled resonance energies and widths, focused on high-lying states at selected parameter points — was carried out by Wang, Chen, and Guo [6]. (For Debye screening the combined field–screening problem has a longer nonperturbative lineage [7–9].) What does not exist for the ECSC case is systematic coverage: a resonance map of the combined field–screening plane, the fate of excited Stark manifolds, or a controlled comparison against Debye screening that isolates what the oscillating tail actually does to field ionization. This paper supplies the three.

The tool is the coupled-channel complex-scaling method of our Debye companion study [9]: Eq. (1) is the half-sum of two Yukawa potentials at complex screening parameters $\lambda(1 \pm i)$, so every screened matrix element reuses the Debye code path at complex argument, and the identity yields at once the analytic bound $\theta_c < \pi/4$ on the rotation angle. The engine verifies its hydrogen limit on import — a failed check aborts any run — and its Yukawa mode reproduces the companion engine to 2×10^{-16}.

II. METHOD

The $m = 0$ coupled-channel complex-scaling machinery is that of Ref. [9]; we restate it here so the paper is self-contained. Each angular channel $\ell = 0, \ldots, \ell_{\max}$ carries the orthonormal radial functions

$$\varphi_i^{(\ell)}(r) = N_{i\ell}\, x^{\ell+1} e^{-x/2} L_i^{(2\ell+2)}(x), \qquad x = 2\beta r, \tag{2}$$

with $L_i^{(a)}$ generalized Laguerre polynomials, $N_{i\ell}$ fixed by $\int_0^\infty \varphi_i \varphi_j\, dr = \delta_{ij}$, and a common scale $\beta = 0.35$

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for all channels; production settings are $n_b = 70$ functions per channel, $\ell_{\max} = 14$ (1050 degrees of freedom), and a shared 3000-point Gauss–Legendre quadrature on $x \in [0, 760]$ (basis-overlap identity certified to 1.3×10^{-12} on every constructed basis, so the problem is a standard complex-symmetric eigenproblem). Adjacent channels couple through the dipole blocks $(\ell+1)/\sqrt{(2\ell+1)(2\ell+3)}\,\langle\varphi_i|r|\varphi_j\rangle$, and every operator is scaled analytically [10–14]; the Stark term is covered by the Herbst–Simon theory [15, 16]. The single ECSC-specific element is the screened factor: writing $e^{-\lambda r}\cos\lambda r = \frac{1}{2}[e^{-\mu_+ r} + e^{-\mu_- r}]$ with $\mu_\pm = \lambda(1 \pm i)$, the complex-scaled screening enters as two Yukawa evaluations at $\mu_\pm r e^{i\theta_c}$. Decay of these factors requires $\mathrm{Re}(\mu_\pm e^{i\theta_c}) > 0$, i.e.

$$\theta_c < \pi/4, \tag{3}$$

a hard analytic cap (enforced in the code) comfortably above the production angles $\theta_c = 0.18/0.28$ (retry 0.30/0.42). Resonances are identified by parametric continuation: at each new λ the search window is centered on $E_{\mathrm{pred}} = E_b^{F=0}(\lambda) + [E(\lambda_{\mathrm{prev}}) - E_b^{F=0}(\lambda_{\mathrm{prev}})]$ while the bound level exists, and on complex linear extrapolation beyond; the eigenvalue nearest the prediction at $\theta_c = 0.18$ is paired with the eigenvalue nearest *it* at 0.28, and the pair is accepted if the modulus difference — recorded as the spread — is below a width-scaled tolerance. (θ_c-trajectories over $[0.14, 0.50]$ are computed for the least favorable points.) Uncertainties quoted are these spreads — error estimates calibrated against the anchors below, not bounds.

Verification. The hydrogen limit reproduces $-1/2n^2$ to 7×10^{-11} — the check runs at engine import and a failure raises, so no production run can proceed on a failed assembly; the $\lambda = 0$ row of the production map reproduces the pure-Stark anchors of Ref. [9] (themselves validated to ten digits against the classical Stark values [17–19]); the small-λ expansion of the ground level follows the exact law $E_{1s} = -\frac{1}{2} + \lambda - \lambda^3\langle r^2\rangle/3 + O(\lambda^4)$ — the λ^2 term is absent for ECSC, unlike Debye screening — to the expected order (deviation 1.2×10^{-8} at $\lambda = 0.01$). For the critical screenings, a 60-digit benchmark has recently become available [20]: our $\lambda_c(2p) = 0.148205$ agrees with its $0.148205032\ldots$ on all quoted digits, and our variational $\lambda_c(1s) = 0.72027$ sits within the documented threshold-diffusivity bias [21] of the exact $0.720524085\ldots$ [3, 20]. The tables of Ref. [6] concern high-lying resonances and contain no ground-state data overlapping the present map; a comparison at their excited-state points is planned once their tabulation is at hand.

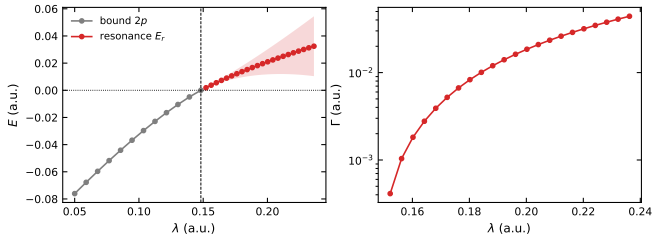


FIG. 1. Field-free $2p$ pole under ECSC screening: bound branch (gray), and the shape resonance born at $\lambda_c(2p) = 0.148205$ (red; band $= \pm\Gamma/2$). Right: width on a log scale.

III. RESULTS

A. Field-free pole structure: the intermediate window

Reference [4] leaves open what happens between $\lambda_c = 0.720524$, where the ground state ceases to be bound, and $\lambda_{r+} = 0.90035$, where sharp s-wave shape resonances appear. Parametric continuation addresses this within a precisely stated scope. Tracking the $1s$ pole across λ_c with fine steps, the pole moves smoothly and basis-stably into the lower half plane, but its rotation response degrades monotonically: the two-angle spread grows from 5×10^{-5} at $\lambda = 0.715$ to 10^{-2} at $\lambda = 1.25$, no θ_c-plateau develops, and — the decisive diagnostic — the tracked ratio $(\Gamma/2)/E_r$ converges to $\tan 2\theta_c$ of the rotation angle used, the textbook signature of the discretized rotated-continuum ray [12–14] rather than of an isolated pole. The statement this establishes is a null result *within uniform complex scaling*, which can only expose poles with $(\Gamma/2)/E_r < \tan 2\theta_{\max} \approx 1.1$ at the ECSC cap: in that sector, no isolated $1s$ resonance survives in the intermediate window — the pole merges into the continuum. A true second-sheet virtual state is invisible to the method by construction, so the result is consistent with — though not a proof of — the virtual-state scenario conjectured in Ref. [4]; a definitive second-sheet statement would require exterior scaling or Jost-function continuation. (The sharp tail resonances above λ_{r+} live too close to the rotated ray for the present uniform-scaling setup; we do not attempt to reproduce them and refer to Ref. [4].) The ray-versus-pole criterion itself is standard [12–14]; what we found operationally decisive near threshold is applying it in ratio form together with λ-insensitivity: any candidate whose width-to-position ratio tracks $\tan 2\theta_c$, or whose parameters do not move with λ, is a discretized ray state, not a resonance.

The $\ell = 1$ story is the clean counterpart (Fig. 1). The $2p$ level unbinds at $\lambda_c(2p) = 0.148205$ — earlier than the Debye value 0.220217 [3, 21], the oscillating tail screens harder at distance — and detaches as a genuine shape resonance behind the centrifugal-plus-tail barrier: near threshold the two-angle spread is at machine level (10^{-13}), and the resonance broadens rapidly as λ grows,

TABLE I. Ground-state resonance parameters (a.u.) at selected screenings. Widths marked $< 10^{-10}$ are below the resolved level. λ^* is the $E_r = 0$ crossing (linear interpolation, ± 0.01). Maximum two-angle spread on the full grid: 1.3×10^{-4}.

F	λ	E_r	Γ	λ^*
0.02	0.3	-0.220638	3.00×10^{-6}	0.686
	0.6	-0.032042	1.43×10^{-2}	
	0.9	$+0.049297$	9.66×10^{-2}	
0.04	0.3	-0.224766	7.00×10^{-4}	0.681
	0.6	-0.034837	3.43×10^{-2}	
	0.9	$+0.066174$	1.35×10^{-1}	
0.06	0.3	-0.231777	5.27×10^{-3}	0.681
	0.6	-0.037618	5.62×10^{-2}	
	0.9	$+0.076982$	1.68×10^{-1}	
0.08	0.3	-0.240144	1.54×10^{-2}	0.681
	0.6	-0.040327	7.95×10^{-2}	
	0.9	$+0.085034$	1.99×10^{-1}	
0.10	0.3	-0.248664	3.04×10^{-2}	0.682
	0.6	-0.042906	1.04×10^{-1}	
	0.9	$+0.091514$	2.30×10^{-1}	

reaching $\Gamma \sim 5 \times 10^{-2}$ by $\lambda = 0.24$.

B. The ground-state (F, λ) map: through the threshold

Figure 2 shows the ground-state resonance on the grid $F \in \{0.02, \ldots, 0.10\} \times \lambda \in \{0, 0.05, \ldots, 0.90\}$: all 95 points tracked by continuation with zero losses, two-angle spreads below 1.5×10^{-4} at the broadest corner and below 10^{-7} over 58% of the grid (machine-readable tables provided). Table I lists a subset.

The headline is the left panel: at every field strength the ground resonance crosses $E_r = 0$ at $\lambda^* \approx 0.68$ (from 0.686 at $F = 0.02$ to 0.681–0.682 at stronger fields, a shallow minimum near $F = 0.06$; total variation below 0.01) and continues smoothly to positive energies, reaching $E_r = +0.049$, $\Gamma = 0.097$ at $F = 0.02$, $\lambda = 0.9$. The near field-independence of λ^* is a corollary of the center panel: the field-induced displacement collapses toward zero as $\lambda \to \lambda^*$ for every F, so the crossing location is pinned by the screening alone. The contrast with the field-free limit is structural: for $F \neq 0$ the complex-scaled Stark operator has empty essential spectrum [15, 16] — there is no continuum ray at threshold to absorb the pole, every eigenvalue is a discrete resonance, and the crossing is guaranteed smooth — whereas at $F = 0$ the same pole merges into the rotated continuum within our method's resolution (Sec. III A). A probe of the connection between the two regimes at the $\lambda = 0.90$–0.95 edge makes this concrete: tracking the resonance down in field, it remains cleanly isolated (spreads 10^{-7}–4×10^{-9}) to $F \approx 0.003$, with $(\Gamma/2)/E_r$ climbing from 0.98 toward the exposure boundary $\tan 2\theta_{\max} \approx 1.1$, below which tracking degrades — the pole exits the sector uniform scaling can expose as the field is removed, consistently with the field-free merging and with no sign, along this path, of connecting to a narrow tail resonance. On the Debye side, the companion map [9] never reaches such a crossing — but that comparison is at equal λ, which Sec. III C argues is not the right variable: in units of the respective critical screenings the ECSC crossing sits at $\lambda^*/\lambda_c \approx 0.94$, while the Debye analog would require $\lambda \approx \lambda_c = 1.19$, far beyond the mapped range. The physical statement is that the ECSC scale compression brings the positive-energy–ground regime — tabulated for Debye screening only at extreme parameters [7] — down into the moderate-screening window.

Two secondary features mirror the Debye map [9] and one does not. Screening again catalyzes field ionization (right panel: at $F = 0.02$ the width climbs eight orders of magnitude across the grid), and the perturbative Stark regime again shrinks with screening. The non-mirroring feature is the field displacement (center panel): it turns *positive* as $\lambda \to \lambda_c$, i.e. near the critical screening the field *raises* the resonance position relative to the vanishing bound level — a threshold effect absent from the Debye map at comparable screenings.

C. Debye versus ECSC at matched binding: what the oscillating tail does to ionization

The comparison that isolates the tail effect must remove the trivial difference — at equal λ the two potentials bind differently. We therefore match the *binding*: for each anchor E_b we solve for λ_{Y} (Yukawa) and λ_{E} (ECSC) such that the field-free ground level equals E_b to 10^{-9}, then compare the Stark widths at equal F. Table II shows the result: at matched binding the ECSC width is systematically *smaller* — by about a factor of two at $F = 0.04$ ($\Gamma_{\mathrm{E}}/\Gamma_{\mathrm{Y}} = 0.45$–$0.57$) and less at $F = 0.08$ (0.65–0.82). The quasistatic exponent is identical by construction, so the suppression is a pure barrier-shape effect, and its origin is the cosine decay *before* the first zero: at the $E_b = -0.30$ anchor the entire tunneling barrier lies inside $r_0 = \pi/2\lambda_{\mathrm{E}} = 7.6$ (downfield barrier peak near $r = 4.4$ at $F = 0.04$, outer turning point near 7.5), and across that region the ratio $V_{\mathrm{ECSC}}/V_{\mathrm{Y}}$ falls from 0.89 at $r = 3$ to 0.15 at $r = 7$ — the cosine factor extinguishes the assisting tail well before changing its sign, so the ECSC barrier is both higher and wider; the sign flip beyond r_0 merely seals the effect. As F grows the barrier peak moves inward (to $r \approx 3.3$ at $F = 0.08$), into the region where the two potentials nearly coincide, and the state approaches the over-barrier regime — so the suppression fades, both trends visible in the table.

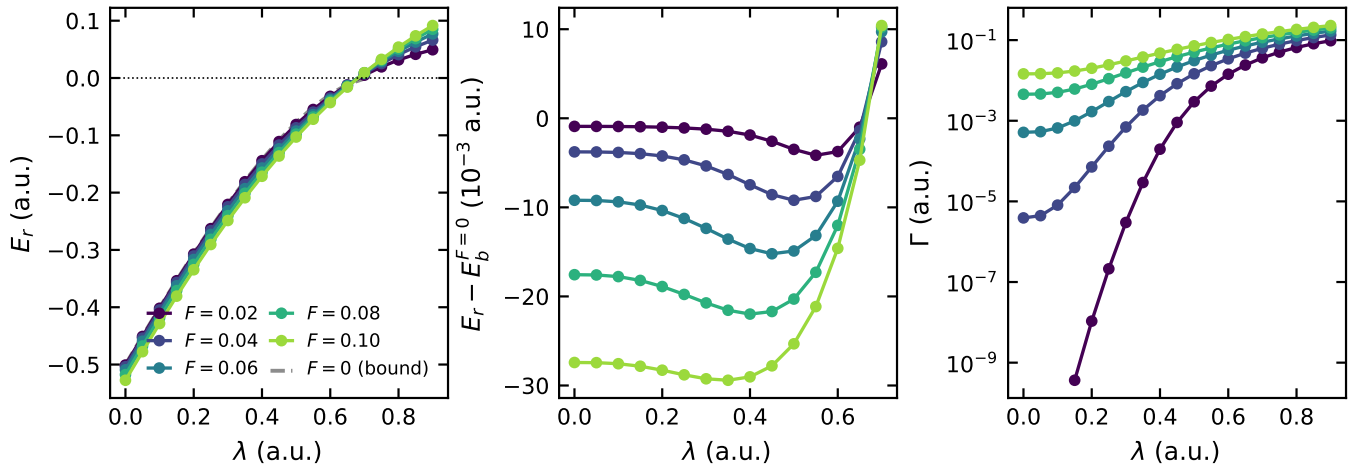


FIG. 2. Ground-state resonance across the (F, λ) plane under ECSC screening. Left: position E_r (dashed: $F = 0$ bound level; dotted: $E_r = 0$). Center: field-induced displacement relative to the bound level (plotted where the $F = 0$ level exists). Right: width (log scale; points shown where Γ exceeds ten times the spread).

TABLE II. Ground Stark widths at matched field-free binding E_b. λ_Y, λ_E: screening parameters at which Yukawa/ECSC give the same E_b (10^{-9} match).

E_b	λ_Y	λ_E	F	Γ_E/Γ_Y
−0.40	0.10823	0.10091	0.04	0.551
			0.08	0.823
−0.30	0.23699	0.20697	0.04	0.447
			0.08	0.677
−0.20	0.39703	0.32394	0.04	0.493
			0.08	0.648
−0.12	0.56482	0.43240	0.04	0.571
			0.08	0.660

D. The $n = 2$ doublet

The $m = 0$ doublet at $F = 0.02$ (parabolic labels $k = \pm 1$ assigned by continuation from $\lambda = 0$, not quantum numbers) behaves as under Debye screening but on a compressed λ scale, consistent with $\lambda_c(2p)$ (Fig. 3): the uphill component crosses $E_r = 0$ at $\lambda \approx 0.093$ (Debye: 0.112 [9]) and continues to $E_r = +0.127$, $\Gamma = 0.079$ at $\lambda = 0.30$; the downhill width roughly triples over the same range ($0.061 \to 0.208$), with the ratio Γ_{-1}/Γ_{+1} drifting $1.97 \to 2.63$. Spreads are below 10^{-5} except at four points (worst 6×10^{-3} for the downhill component at $\lambda = 0.30$, which we quote to two digits only).

E. Physical scope

In the quantum-hydrodynamic derivation [1] λ is the quantum screening wavenumber, so $\lambda = 0.1$–0.9 a.u. corresponds to screening lengths of 1–10 a_0 (0.05–0.5 nm)

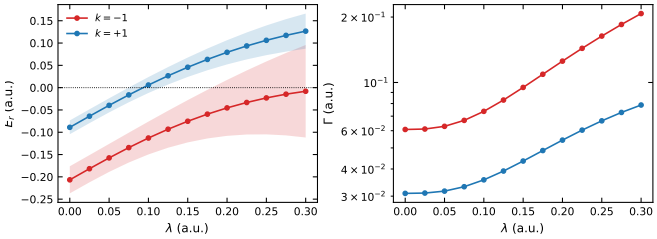


FIG. 3. The $n = 2$, $m = 0$ Stark doublet at $F = 0.02$ under ECSC screening. Left: positions with $\pm\Gamma/2$ bands; k labels the $\lambda = 0$ parabolic character, carried by continuation. Right: widths.

— metallic to warm-dense-matter electron densities, roughly 10^{22}–10^{24} cm^{-3}. The ECSC form is itself a model whose validity at the extremes remains under discussion [5]; the present map is a statement about that model's spectral content, in the same spirit as the Debye-model literature it extends, and the matched-binding comparison of Sec. III C is designed to be the model-robust part of the physics: it isolates a barrier-shape mechanism (tail extinction before the first zero) that any oscillatory screening form shares qualitatively.

IV. CONCLUSIONS

Three results: the intermediate screening window of Ref. [4] is resolved — as a null result within uniform complex scaling, the ground pole merges into the continuum, consistent with the virtual-state scenario; the first systematic (F, λ) resonance map under ECSC screening shows the ground state crossing into a positive-energy resonance at $\lambda^* \approx 0.68$ for every field, pinned by the vanishing field displacement at threshold; and the matched-

binding comparison shows the oscillating tail suppresses field ionization by up to a factor of two, a barrier-shape effect that fades with increasing field. The extension of the Debye machinery that made this possible is a single identity — the ECSC potential as a half-sum of complex-parameter Yukawas, with its $\theta_c < \pi/4$ cap. The machinery extends unchanged to excited manifolds, $m \neq 0$, and other screening families; the relativistic (Dirac) version of the combined problem is the natural next step and is under way.

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